

Name (Pinyin): _____

姓名: _____

S&T2025 Basic C Programming Language Second Mock Test (First Checkpoint)

Time allowed: 1 hour and 30 minutes

INSTRUCTIONS

1. This is a **CLOSED BOOK** examination.
2. This examination paper contains **10 questions** and 8 pages. Answer **all** questions.
3. The full score for each question is as follows:

Question 1: 5	Question 4: 5	Question 7: 10	Question 10: 30
Question 2: 5	Question 5: 5	Question 8: 10	
Question 3: 5	Question 6: 5	Question 9: 20	
4. Answers to the questions are to be written in this booklet.
5. Students are **NOT** allowed to bring any programmable calculator, machine translator, or dictionary with them. If you have any of these items, you are to surrender them before the test starts. **Violation of this regulation will be severely dealt with and may result in an expulsion from the bridging course.**
6. Students are also **NOT** allowed to bring any paper to the Examination Hall.
7. All the working steps must be shown.
8. This test paper will be collected at the end of the examination.

No.	Score 1 (Marker)	Score 2 (Checker)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

1. What will the following program print on the screen or is there a compilation error or otherwise?

```
# include <stdio.h>
void main()
{
    int a=0, b=50, count=-2;

    while (2*a < b)
    {
        a += 2;
        b /= 2;
        ++count;
    }

    printf("a=%d, b=%d, count=%d  ", a, b, count);
}
```

(5 marks)

Answer:

2. What will the following program print on the screen or is there a compilation error or otherwise, if the input from the keyboard is 1 3 2 ?

```
# include <stdio.h>
void main()
{
    int a, b, c;

    printf("Enter a, b, c: ");
    scanf("%d%d%d", &a, &b, &c);

    if (a > b)
    if (b > c)
    printf("X");
    else
    printf("Y");
    else
    if (b > c)
    printf("Z");
}
```

(5 marks)

Answer:

3. What will the following program print on the screen or is there a compilation error or otherwise?

```
#include <stdio.h>
void main()
{
    int a = 30, b = 2, sum = 0;

    while ( (a>40) || (a = b) )
    {
        sum++;
        a++;
    }
    printf ("\n a= %d, b= %d, sum= %d", a, b, sum);
}
```

(5 marks)

Answer:

4. What will the following program print on the screen or is there a compilation error or otherwise?

```
#include <stdio.h>
int main (void)
{
    int i = 2;
    int x = 0;

    for ( ;i <= 14; i+=2)
    {
        x+=2;
        switch (i%5)
        {
            case 0:
            case 1: i=i+1;
            case 2: i++;
                     continue;
            case 3: i-2;
        }
        i++;
    }

    printf("i=%d, x=%d      ", x, i);
    return 0;
}
```

(5 marks)

Answer:

5. What will the following program print on the screen or is there a compilation error or otherwise?

```
#include <stdio.h>

int f (int );
int g (int );

int main (void)
{
    printf("Answer = %d ", g(f(2)) );

    return 0;
}

int f(int y)
{
    if (y==0) return 0;
    if (y < 5) return g(y);
}

int g(int y)
{
    if (y==0) return 1;
    if (y < 3) return f(y-1) + 1;
}
```

(5 marks)

Answer:

6. What will the following program print on the screen or is there a compilation error or otherwise?

```
#include <stdio.h>

main()
{
    int i = -1;
    int j =- 1;
    int k=0, w =1, m;

    m =!( !(i++&& j++&& k++ || --w));
    printf("%d %d %d %d %d ", i, j, k, w, m);

    return 0;
}
```

(5 marks)

Answer:

7. In the floating-point representation to approximate real numbers, we restrict the mantissa by normalization, i.e., the mantissa cannot start with 00 and 11. What will be the short-coming if this restriction is lifted? Explain your answer with an example using 4 bits for mantissa and 4 bits for exponent.

(10 marks)

Answer:

8. Write a complete C program to copy its input to its output, replacing each continuous two or more blanks by a single blank. Three independent screen shots of the execution are as follows:

Test Run 1:

```
Enter a sentence:Write a complete  c          programe.
Write a complete c programe.
```

Test Run 2:

```
Enter a sentence: Write a      program.
Write a program.
```

Test Run 3:

```
Enter a sentence:      Write 3 programs and compile      them.
Write 3 programs and compile them.
```

(10 marks)

Answer:

9. Write a complete C program with the iterative approach to compute the value of the following series with an accuracy of up to 4 decimal places if it is possible.

$$0.5 - \frac{0.5^2}{2} + \frac{0.5^3}{3} - \frac{0.5^4}{4} + \frac{0.5^5}{5} - \frac{0.5^6}{6} + \frac{0.5^7}{7} \dots$$

(20 marks)

Answer:

10. Write a complete C program with the use of random numbers to approximate the area enclosed by the following 4 equations:

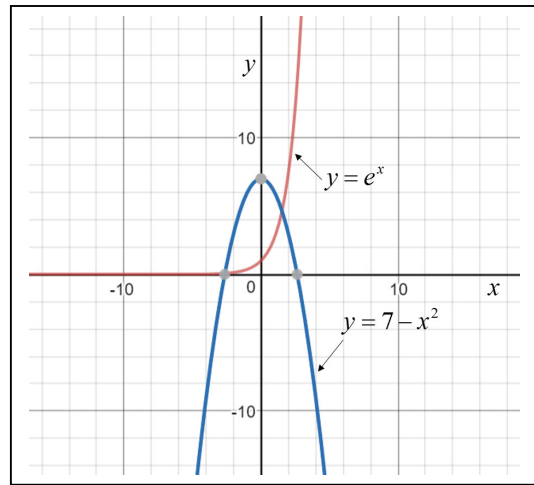
$$y = 7 - x^2$$

$$y = e^x$$

$$x = -2$$

$$x = 1$$

(30 marks)



Answer: